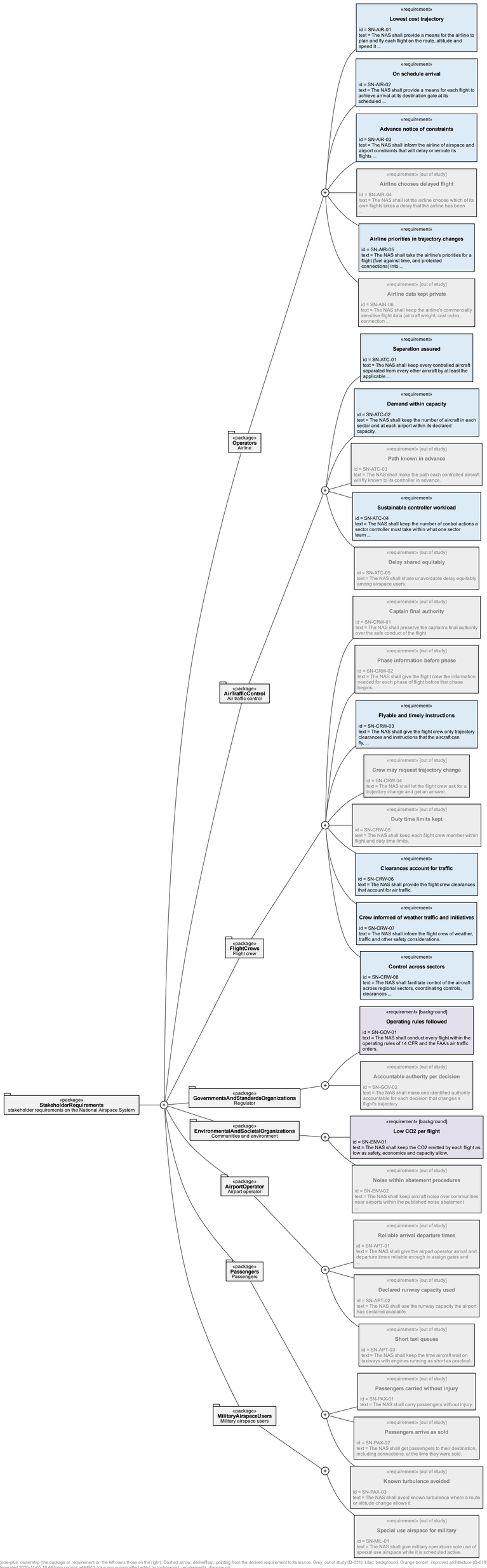
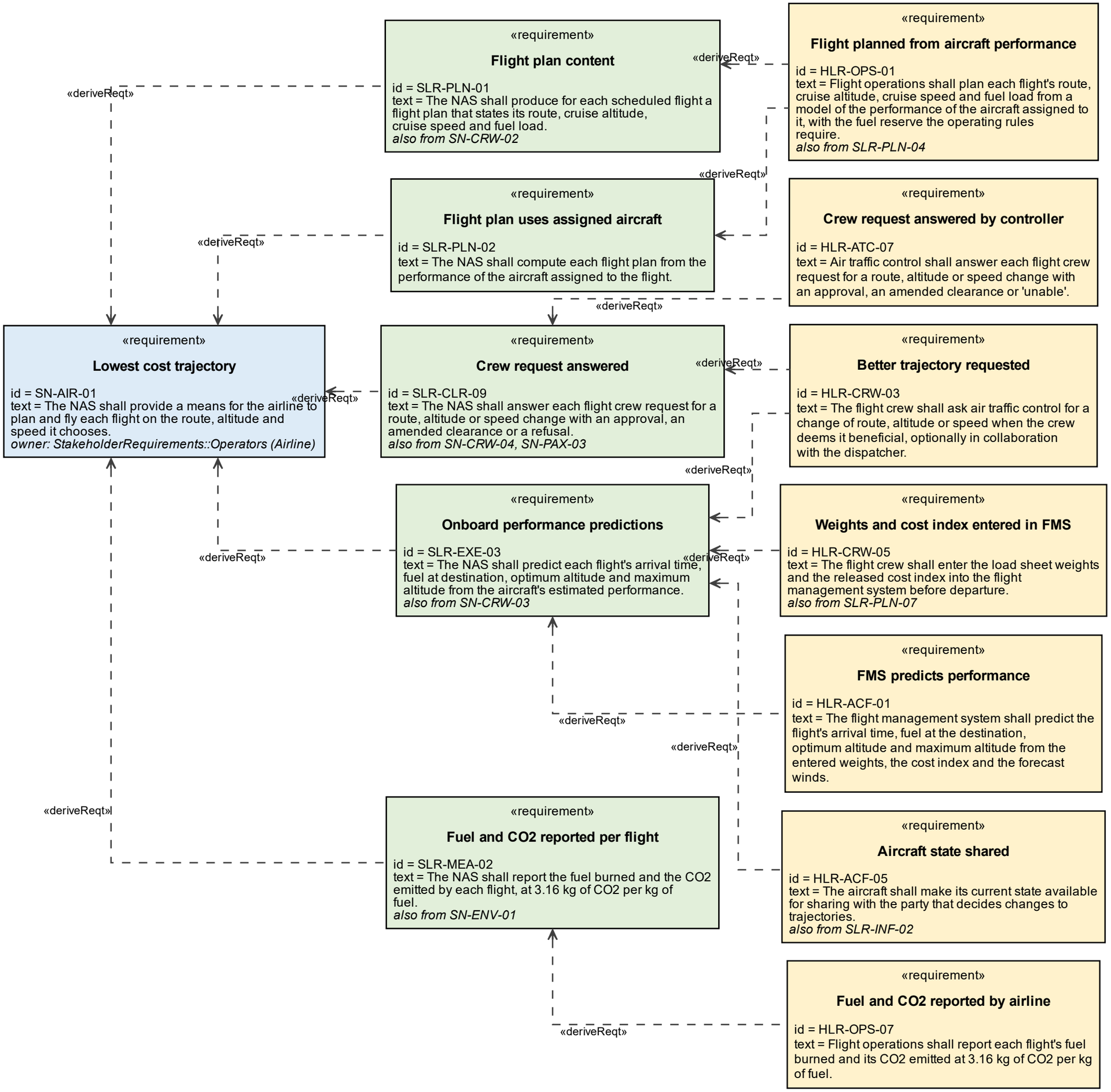
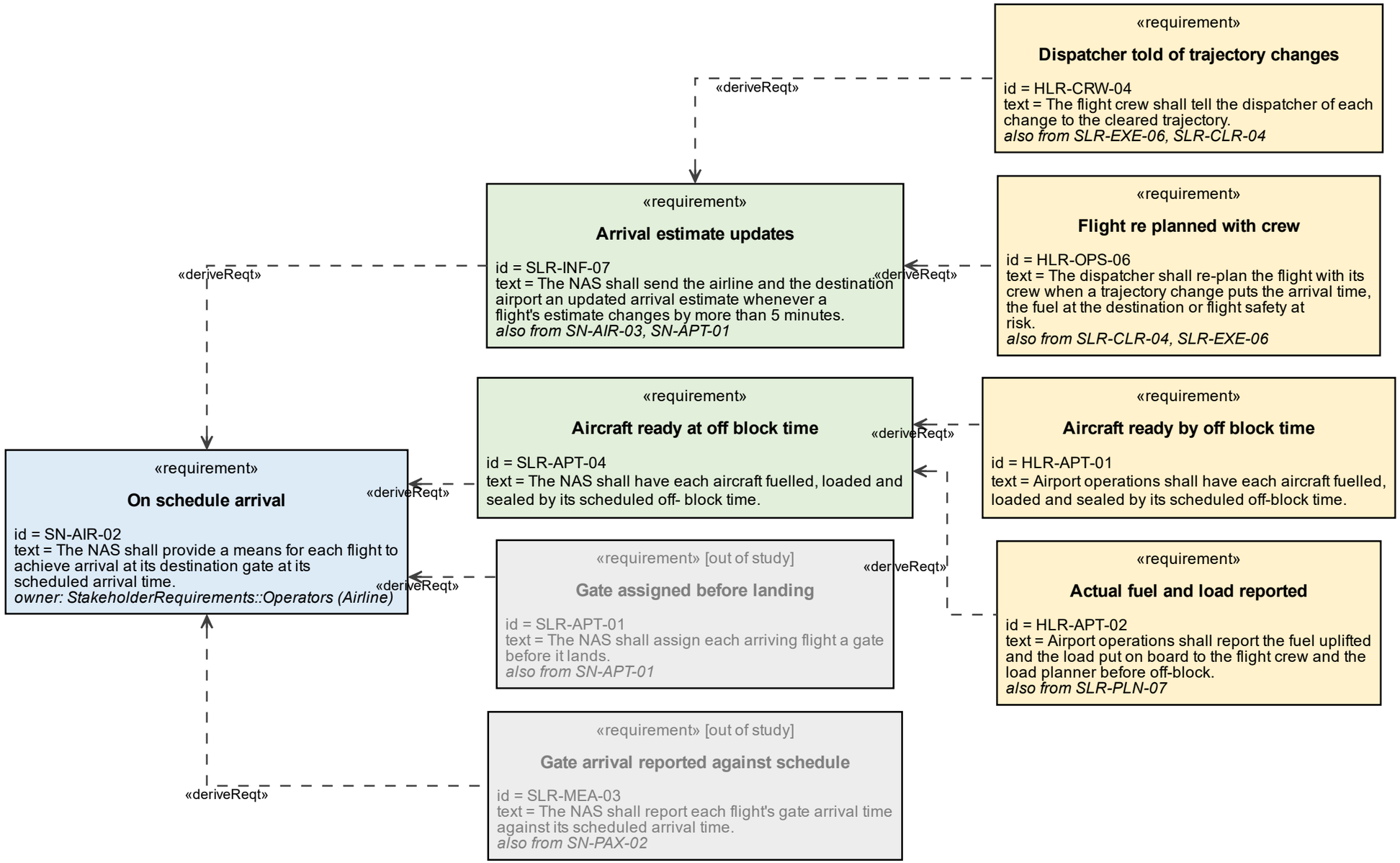


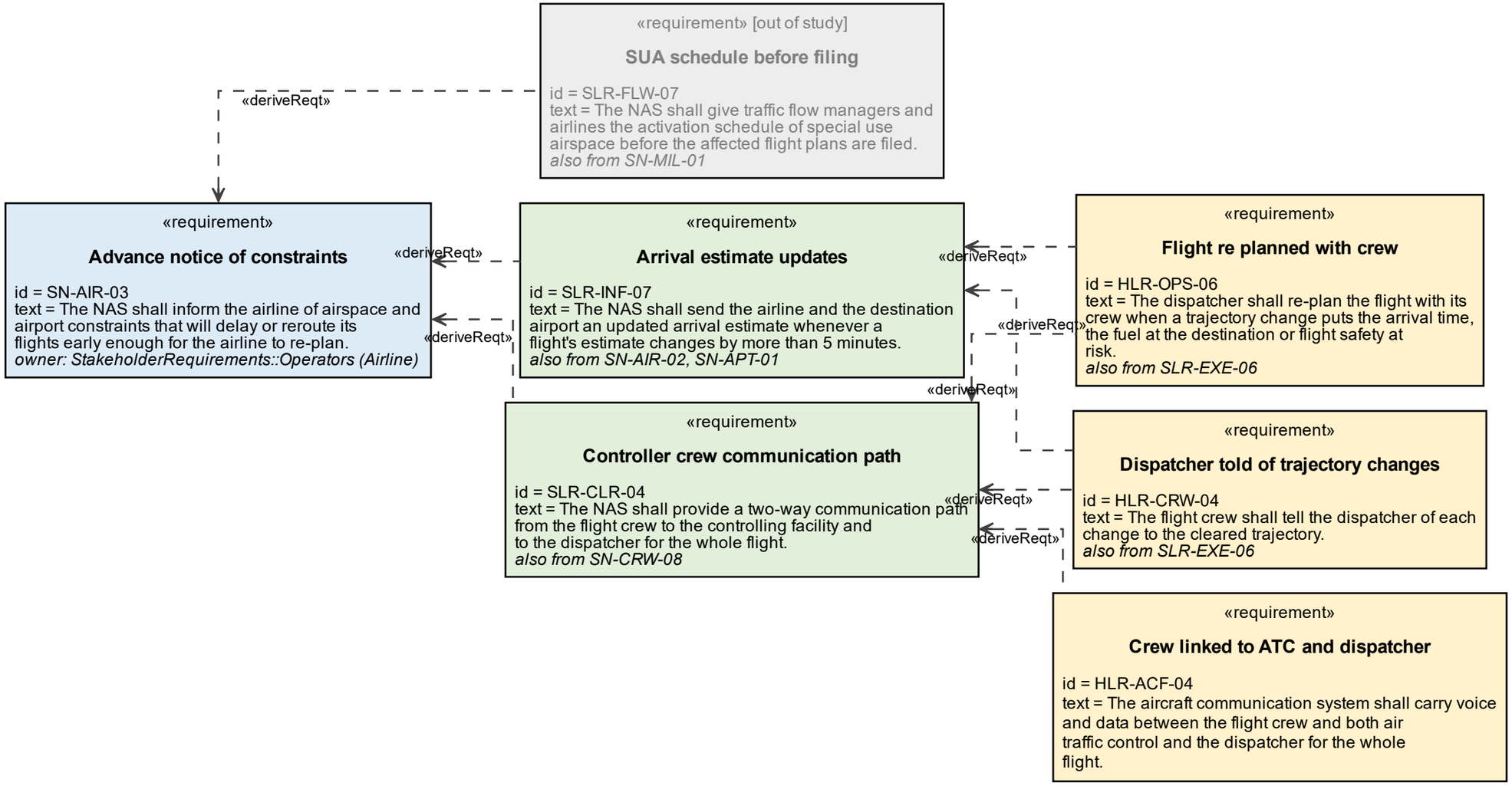
Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.
Requirements: requirements_stakeholders.sysml



Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

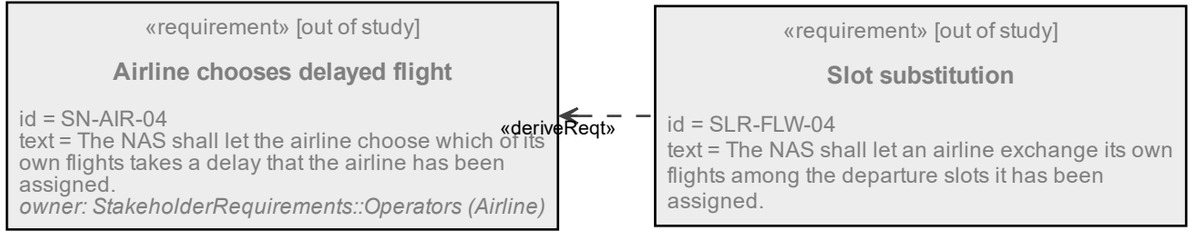


Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

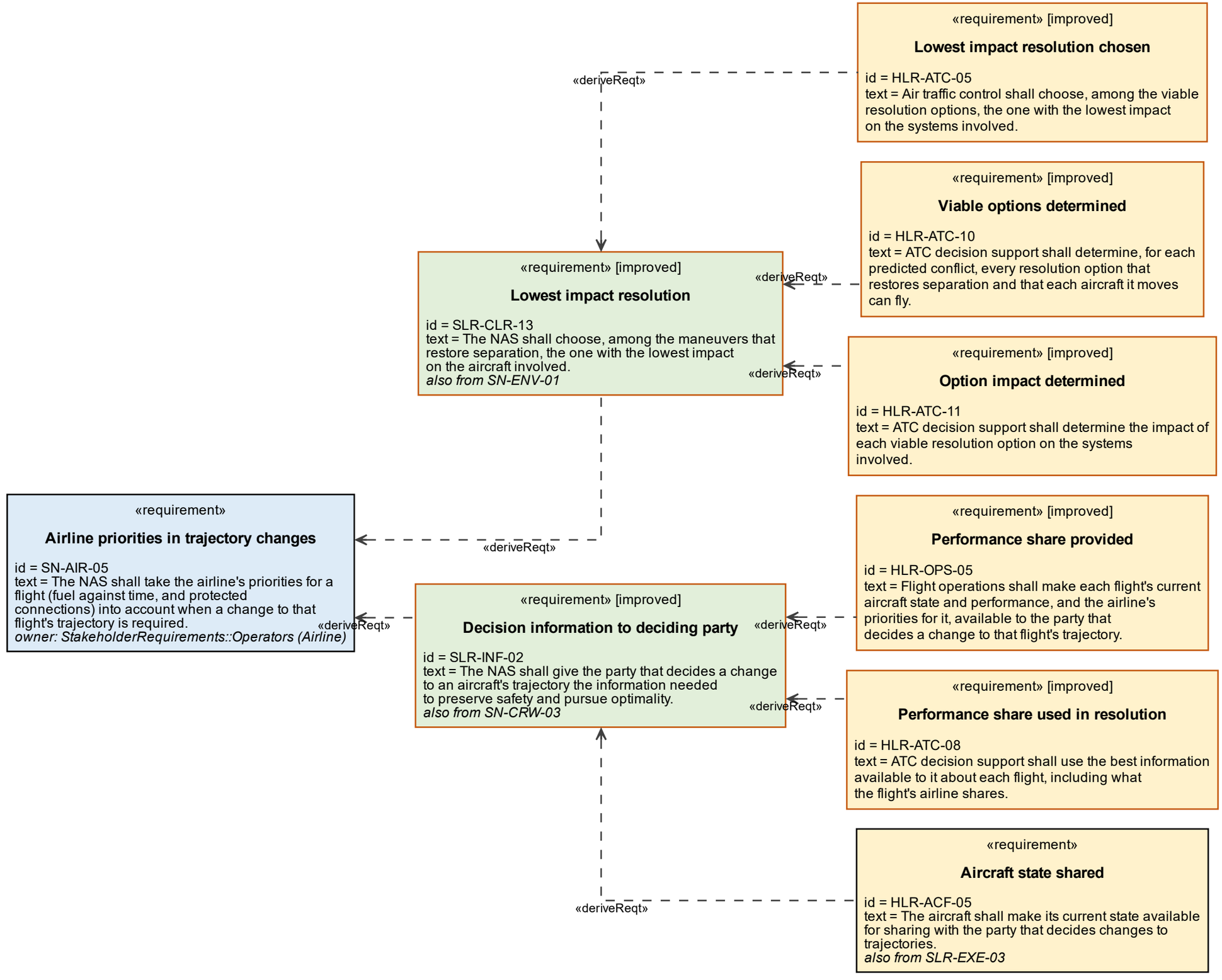


Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

No HLRs: the study is cut at the HLR tier (D-031).

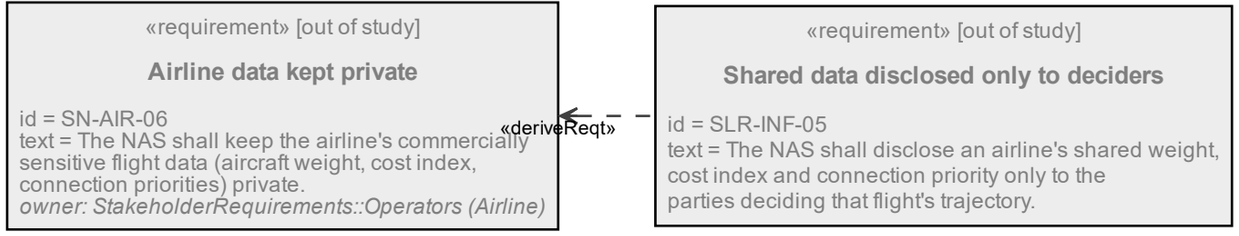


Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.



Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

No HLRs: the study is cut at the HLR tier (D-031).



Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

«requirement»

Flight plan filed by dispatcher

id = HLR-OPS-02
text = The dispatcher shall file each flight's plan with air traffic control before departure.

«deriveReqt»

«requirement»

Flight plan filed with ATC

id = SLR-PLN-06
text = The NAS shall file each flight plan with air traffic control before the flight departs.
also from SN-ATC-02, SN-ATC-03

«deriveReqt»

«requirement»

Flight controlled as cleared

id = HLR-CRW-01
text = The flight crew shall safely control the flight from origin to destination, as cleared by air traffic control with or without negotiation, and in collaboration with flight operations.
also from SLR-CLR-08, SLR-PLN-05

«deriveReqt»

«requirement»

Cleared path held by autoflight

id = HLR-ACF-02
text = The auto flight control system shall hold the aircraft on its cleared route, altitude and speed within the tolerances of the airspace.

«deriveReqt»

«requirement»

Conformance to cleared path

id = SLR-EXE-02
text = The NAS shall fly each aircraft along its cleared route and at its cleared altitude and speed within the navigation and altitude tolerances of the airspace.
also from SN-ATC-03

«deriveReqt»

«requirement»

Separation kept with conformance margin

id = HLR-ATC-01
text = Air traffic control shall keep each pair of aircraft it controls separated by at least the applicable minimum, allowing for each aircraft's permitted deviation from its cleared route, altitude and speed.

«deriveReqt»

«requirement»

Separation minima maintained

id = SLR-CLR-01
text = The NAS shall keep each pair of controlled aircraft separated by at least the minimum that JO 7110.65BB sets for the airspace they are in.
also from SN-CRW-06, SN-PAX-01

«deriveReqt»

«requirement»

Conflict resolved with flyable clearance

id = HLR-ATC-04
text = Air traffic control shall resolve each predicted conflict with a clearance that every aircraft it moves can fly.
also from SLR-CLR-10

«deriveReqt»

«requirement»

Separation assured

id = SN-ATC-01
text = The NAS shall keep every controlled aircraft separated from every other aircraft by at least the applicable separation minimum.
owner: StakeholderRequirements::AirTrafficControl (Air traffic control)

«deriveReqt»

«requirement»

Conflicts predicted

id = SLR-CLR-02
text = The NAS shall predict a loss of separation between controlled aircraft before it occurs.
also from SN-CRW-06

«deriveReqt»

«requirement»

ADS-B state broadcast

id = HLR-ACF-06
text = The aircraft's surveillance system shall broadcast the aircraft's position, altitude, velocity and identity by ADS-B Out.

«deriveReqt»

«requirement» [out of study]

Position altitude identity to facility

id = SLR-CLR-03
text = The NAS shall give the controlling facility the position, altitude and identity of each controlled aircraft.
also from SN-ATC-03

«deriveReqt»

«requirement»

Controller alerted to predicted conflict

id = HLR-ATC-02
text = ATC decision support shall alert the sector controller to each predicted loss of separation before it occurs.

«deriveReqt»

«requirement» [out of study]

Change only by clearance

id = SLR-CLR-05
text = The NAS shall change a controlled aircraft's cleared route, altitude or speed only by a clearance from the facility that has control of that aircraft.
also from SN-GOV-02

«deriveReqt»

«requirement»

Filed intent to sector controller

id = SLR-INF-01
text = The NAS shall give the sector controller the filed route, requested altitude and requested speed of each aircraft before it enters the sector.
also from SN-ATC-03

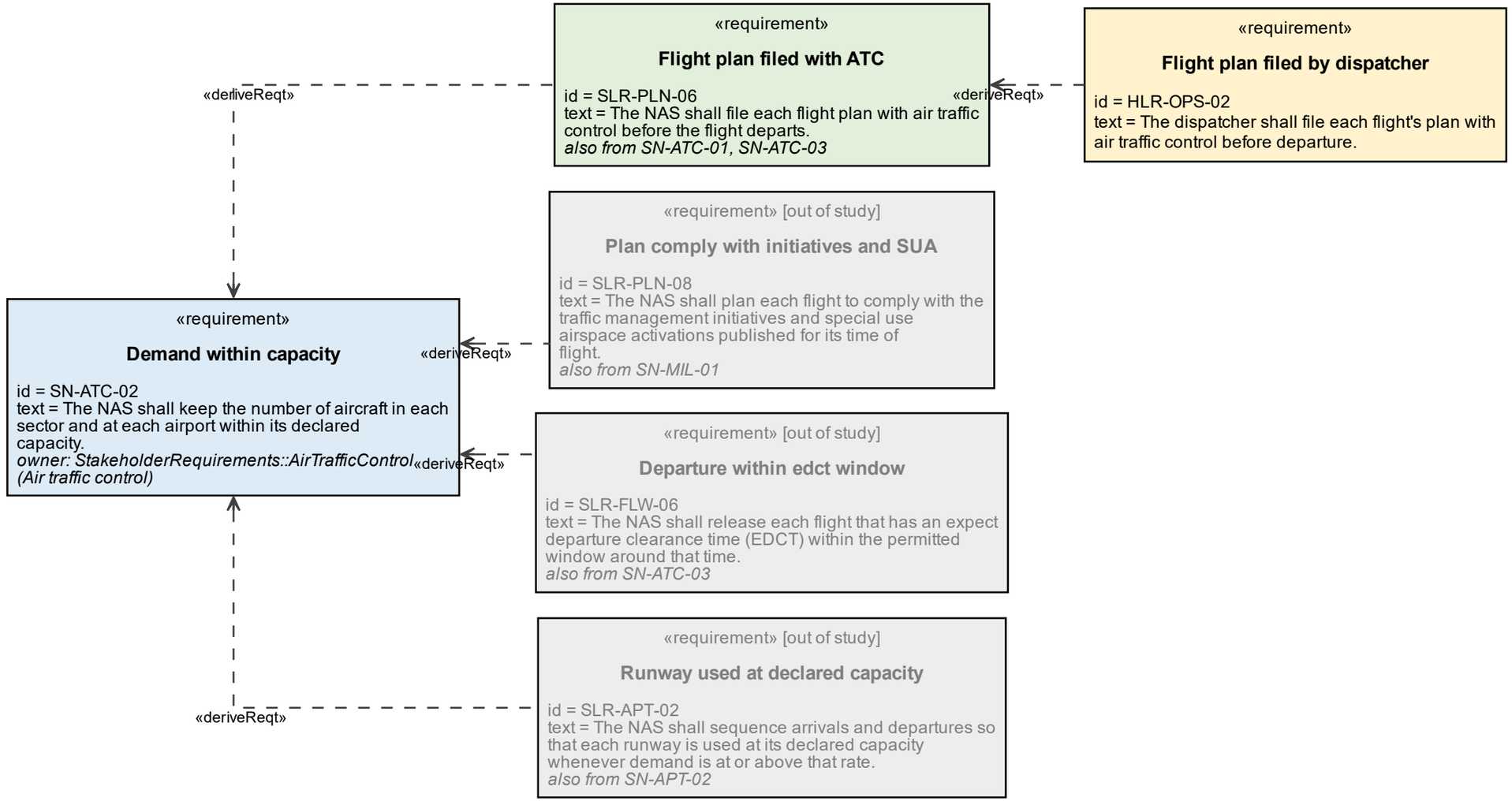
«deriveReqt»

«requirement»

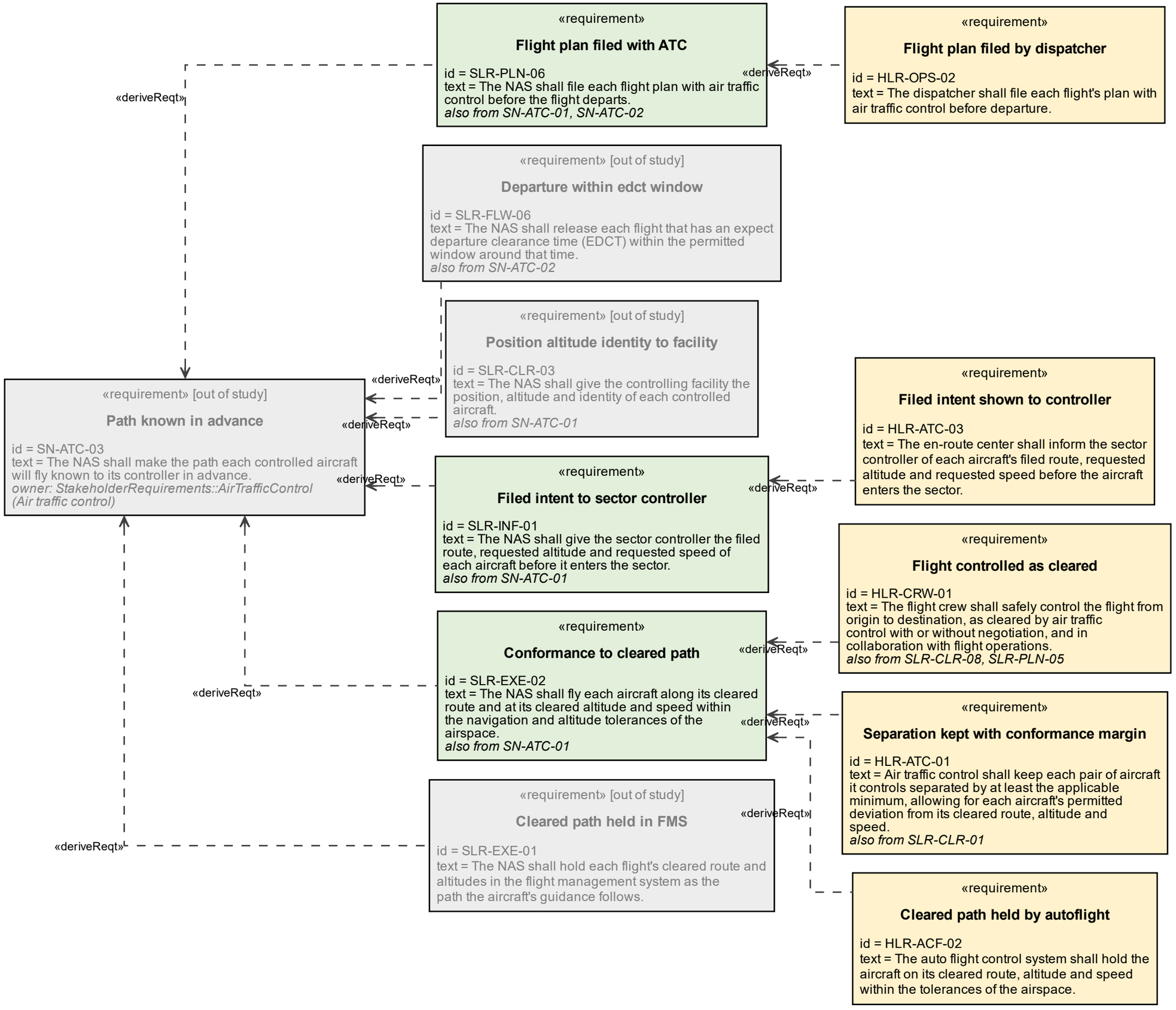
Filed intent shown to controller

id = HLR-ATC-03
text = The en-route center shall inform the sector controller of each aircraft's filed route, requested altitude and requested speed before the aircraft enters the sector.

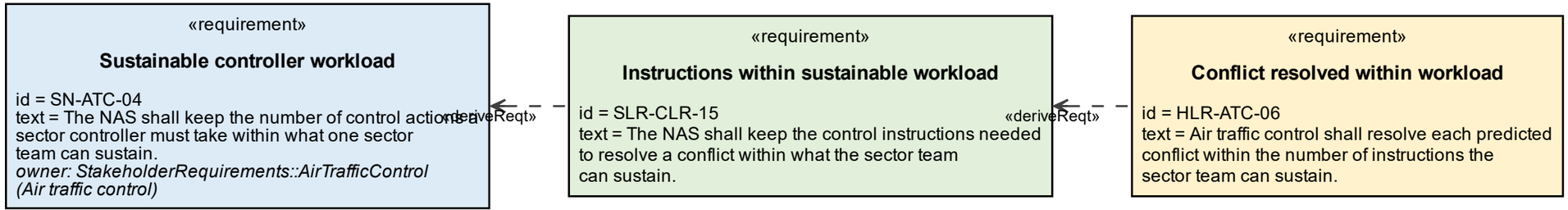
Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.



Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

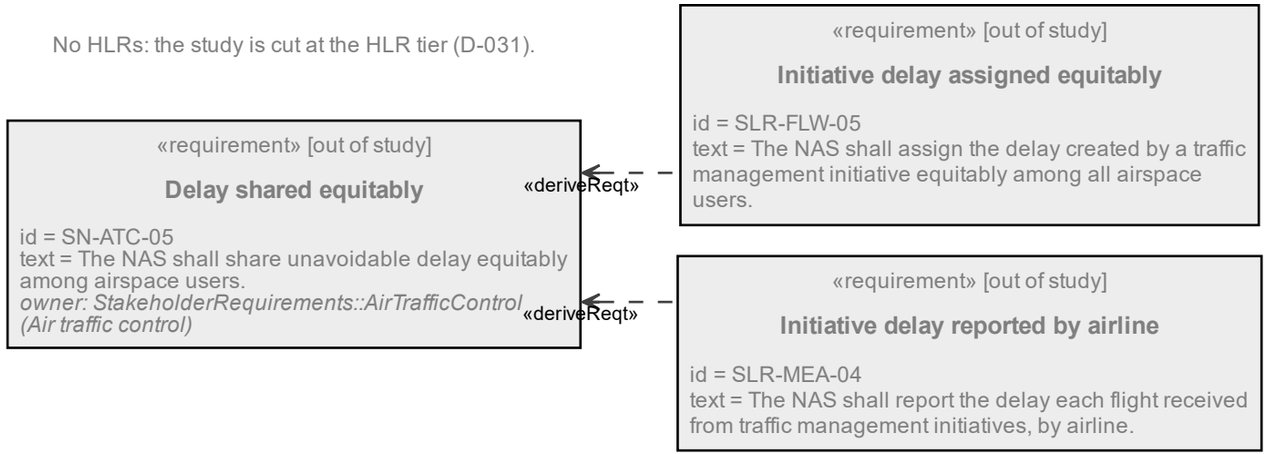


Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

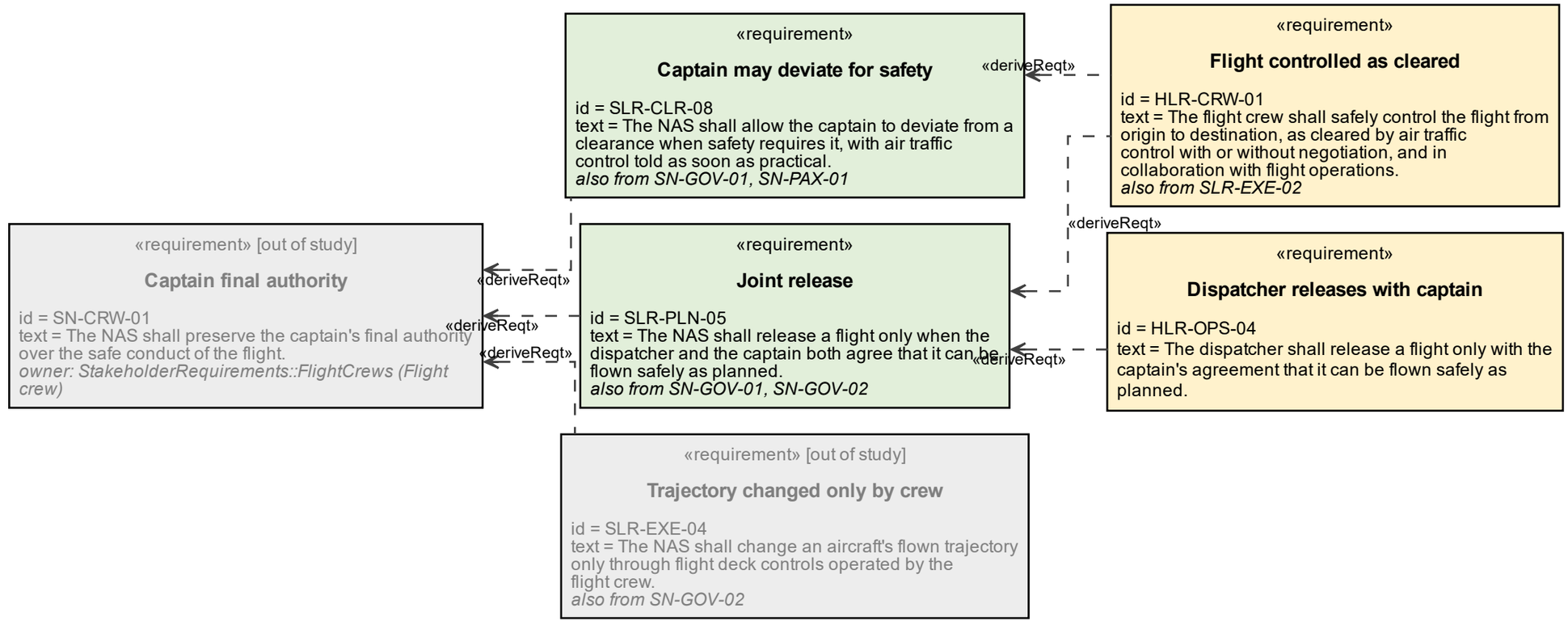


Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

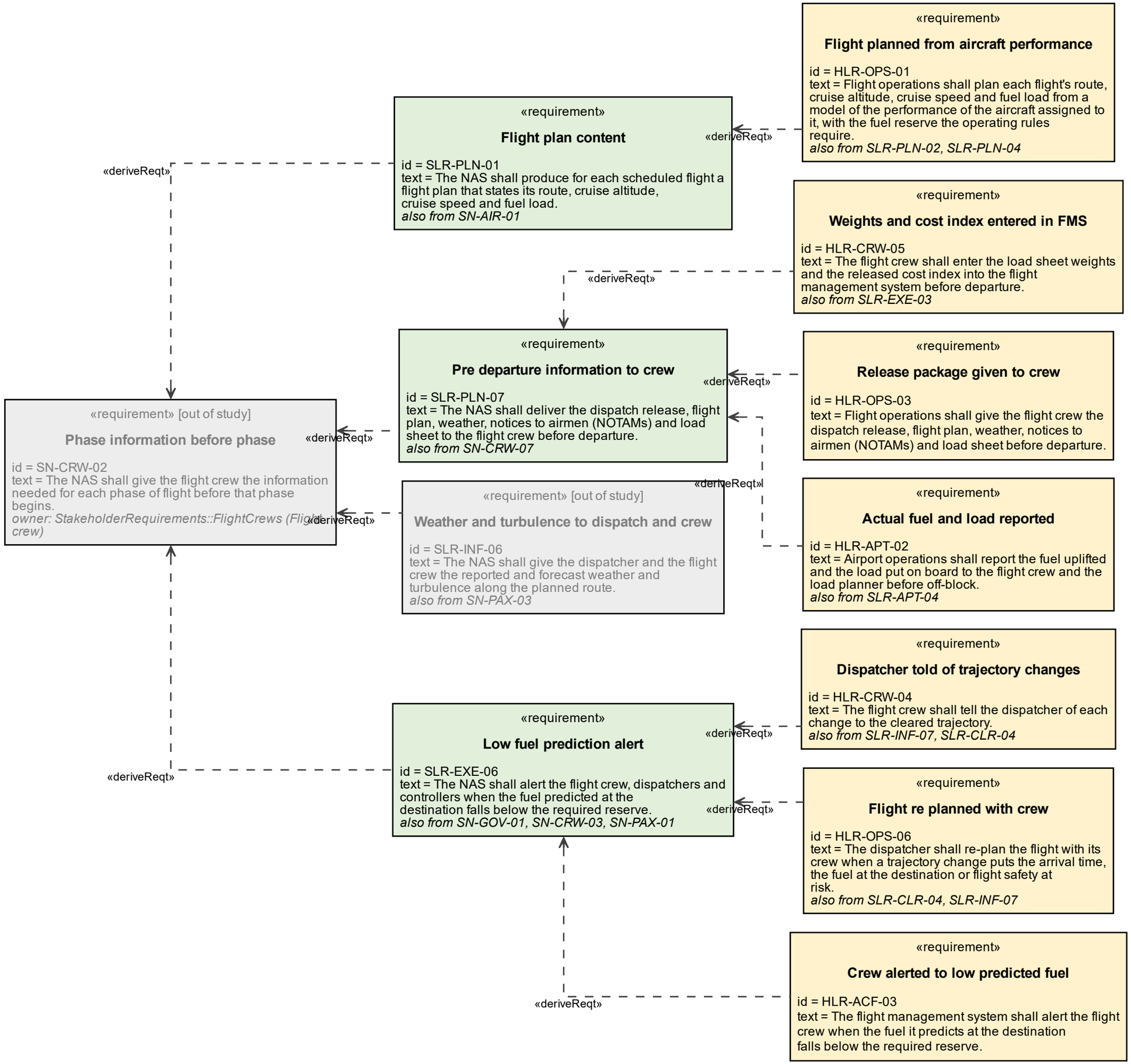
No HLRs: the study is cut at the HLR tier (D-031).



Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.



Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.



Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018). Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

«requirement»

Unflyable clearance refused

id = HLR-CRW-02
text = The flight crew shall accept a clearance only if the aircraft can fly it, and shall answer 'unable' when it cannot.

«deriveReqt»

«requirement»

Clearance within aircraft capability

id = SLR-CLR-10
text = The NAS shall issue only clearances that the aircraft can fly.

«requirement»

Conflict resolved with flyable clearance

id = HLR-ATC-04
text = Air traffic control shall resolve each predicted conflict with a clearance that every aircraft it moves can fly.
also from SLR-CLR-01

«deriveReqt»

«deriveReqt»

«requirement» [improved]

Performance share provided

id = HLR-OPS-05
text = Flight operations shall make each flight's current aircraft state and performance, and the airline's priorities for it, available to the party that decides a change to that flight's trajectory.

«deriveReqt»

«requirement» [improved]

Performance share used in resolution

id = HLR-ATC-08
text = ATC decision support shall use the best information available to it about each flight, including what the flight's airline shares.

«deriveReqt»

«requirement» [improved]

Decision information to deciding party

id = SLR-INF-02
text = The NAS shall give the party that decides a change to an aircraft's trajectory the information needed to preserve safety and pursue optimality.
also from SN-AIR-05

«deriveReqt»

«requirement»

Aircraft state shared

id = HLR-ACF-05
text = The aircraft shall make its current state available for sharing with the party that decides changes to trajectories.

«deriveReqt»

«deriveReqt»

«requirement»

Flyable and timely instructions

id = SN-CRW-03
text = The NAS shall give the flight crew only trajectory clearances and instructions that the aircraft can fly, with enough time to set them up.
owner: StakeholderRequirements::FlightCrews (Flight crew)

«deriveReqt»

«requirement»

Onboard performance predictions

id = SLR-EXE-03
text = The NAS shall predict each flight's arrival time, fuel at destination, optimum altitude and maximum altitude from the aircraft's estimated performance.
also from SN-AIR-01

«deriveReqt»

«requirement»

Better trajectory requested

id = HLR-CRW-03
text = The flight crew shall ask air traffic control for a change of route, altitude or speed when the crew deems it beneficial, optionally in collaboration with the dispatcher.
also from SLR-CLR-09

«deriveReqt»

«requirement»

Weights and cost index entered in FMS

id = HLR-CRW-05
text = The flight crew shall enter the load sheet weights and the released cost index into the flight management system before departure.
also from SLR-PLN-07

«deriveReqt»

«deriveReqt»

«requirement»

FMS predicts performance

id = HLR-ACF-01
text = The flight management system shall predict the flight's arrival time, fuel at the destination, optimum altitude and maximum altitude from the entered weights, the cost index and the forecast winds.

«deriveReqt»

«requirement»

Low fuel prediction alert

id = SLR-EXE-06
text = The NAS shall alert the flight crew, dispatchers and controllers when the fuel predicted at the destination falls below the required reserve.
also from SN-GOV-01, SN-PAX-01, SN-CRW-02

«deriveReqt»

«requirement»

Dispatcher told of trajectory changes

id = HLR-CRW-04
text = The flight crew shall tell the dispatcher of each change to the cleared trajectory.
also from SLR-INF-07, SLR-CLR-04

«deriveReqt»

«requirement»

Flight re planned with crew

id = HLR-OPS-06
text = The dispatcher shall re-plan the flight with its crew when a trajectory change puts the arrival time, the fuel at the destination or flight safety at risk.
also from SLR-CLR-04, SLR-INF-07

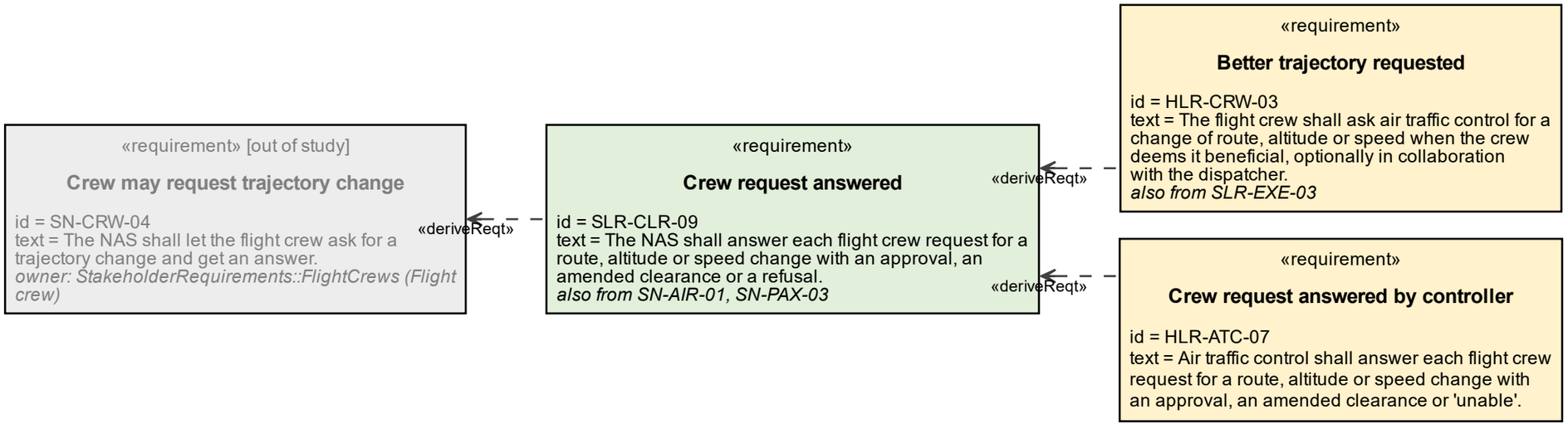
«deriveReqt»

«requirement»

Crew alerted to low predicted fuel

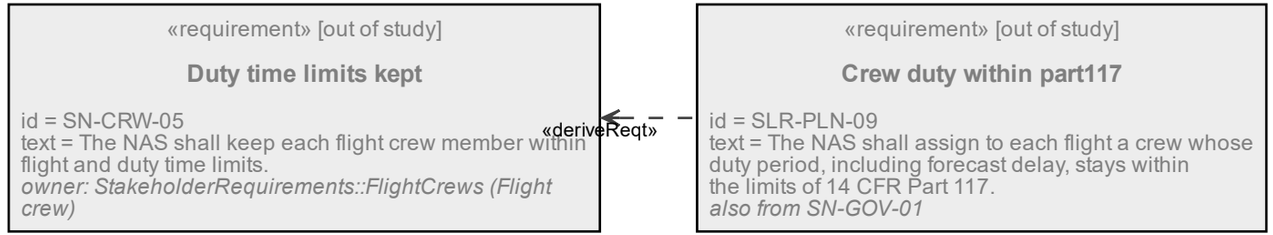
id = HLR-ACF-03
text = The flight management system shall alert the flight crew when the fuel it predicts at the destination falls below the required reserve.

Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

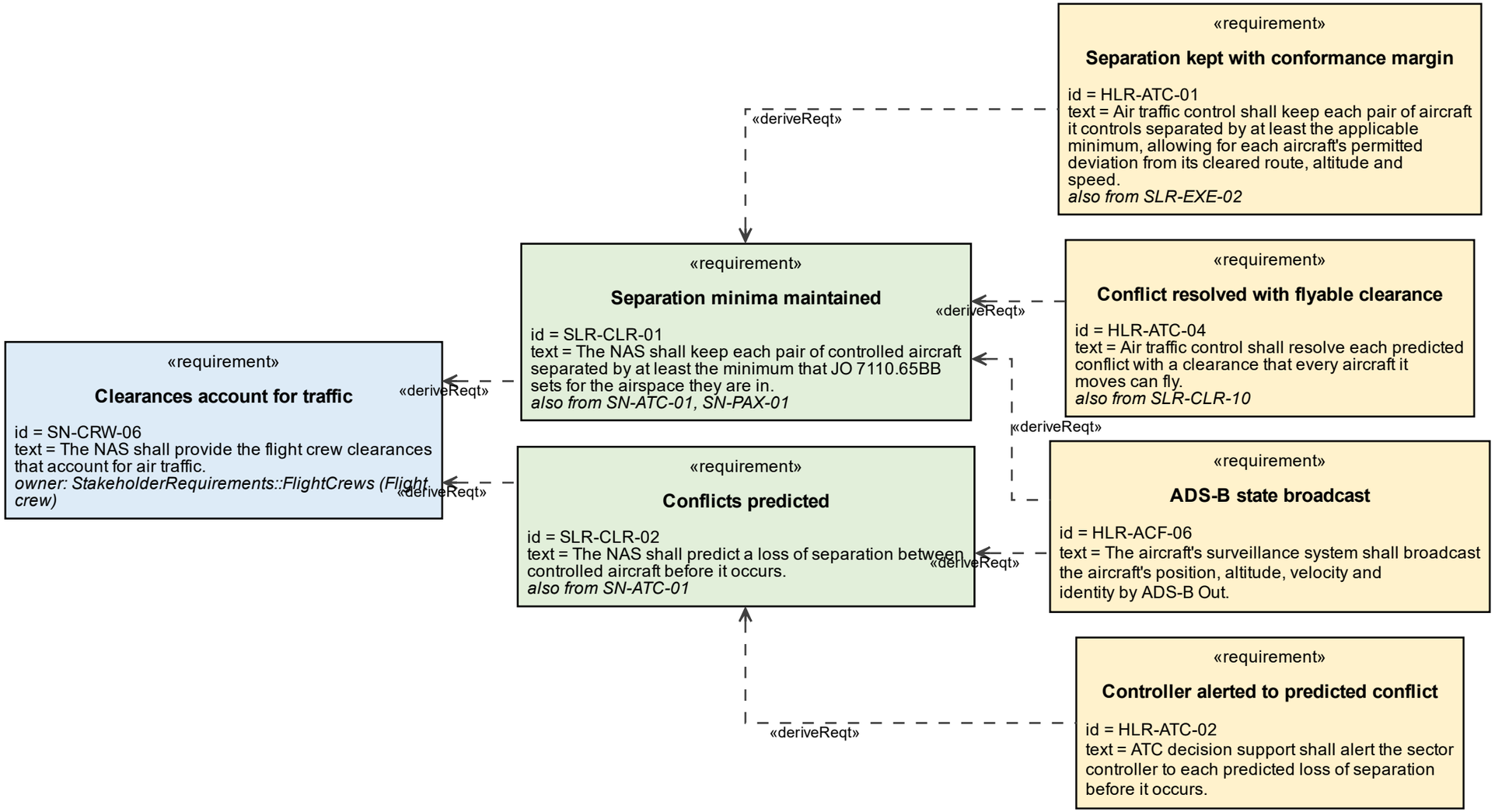


Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

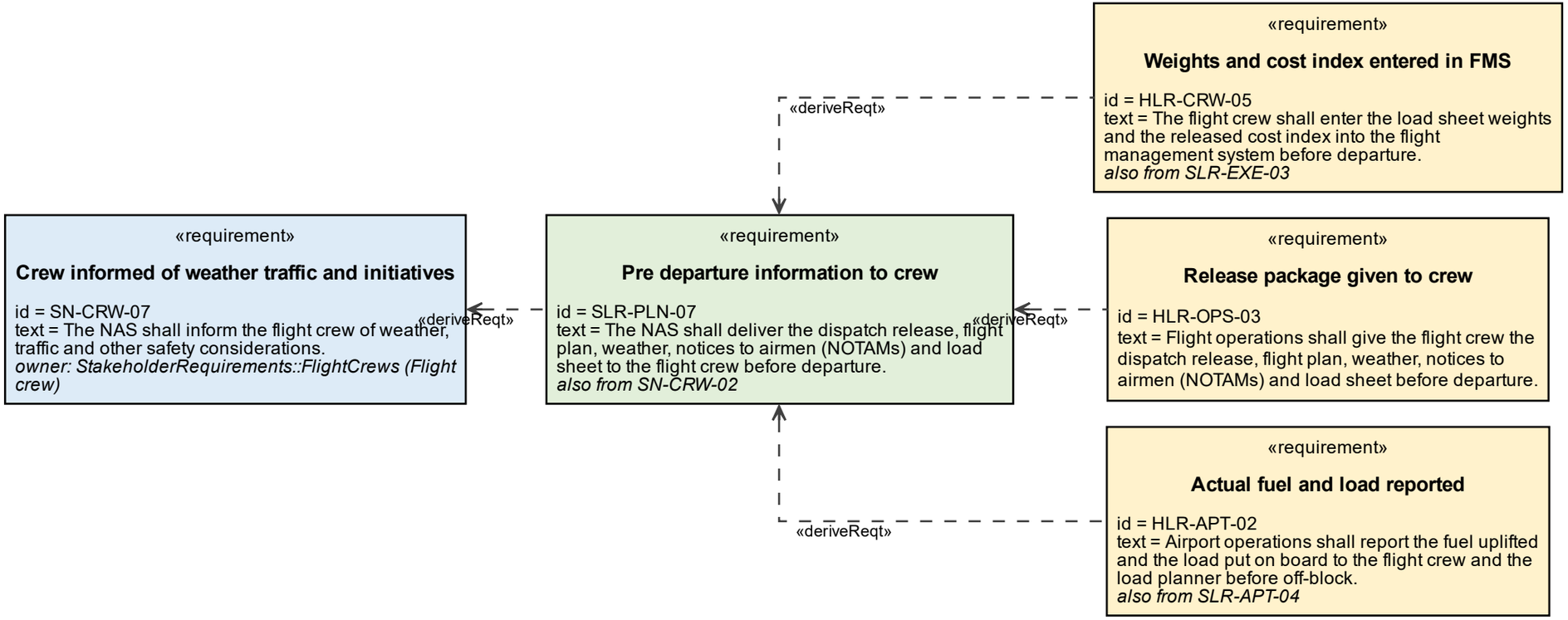
No HLRs: the study is cut at the HLR tier (D-031).



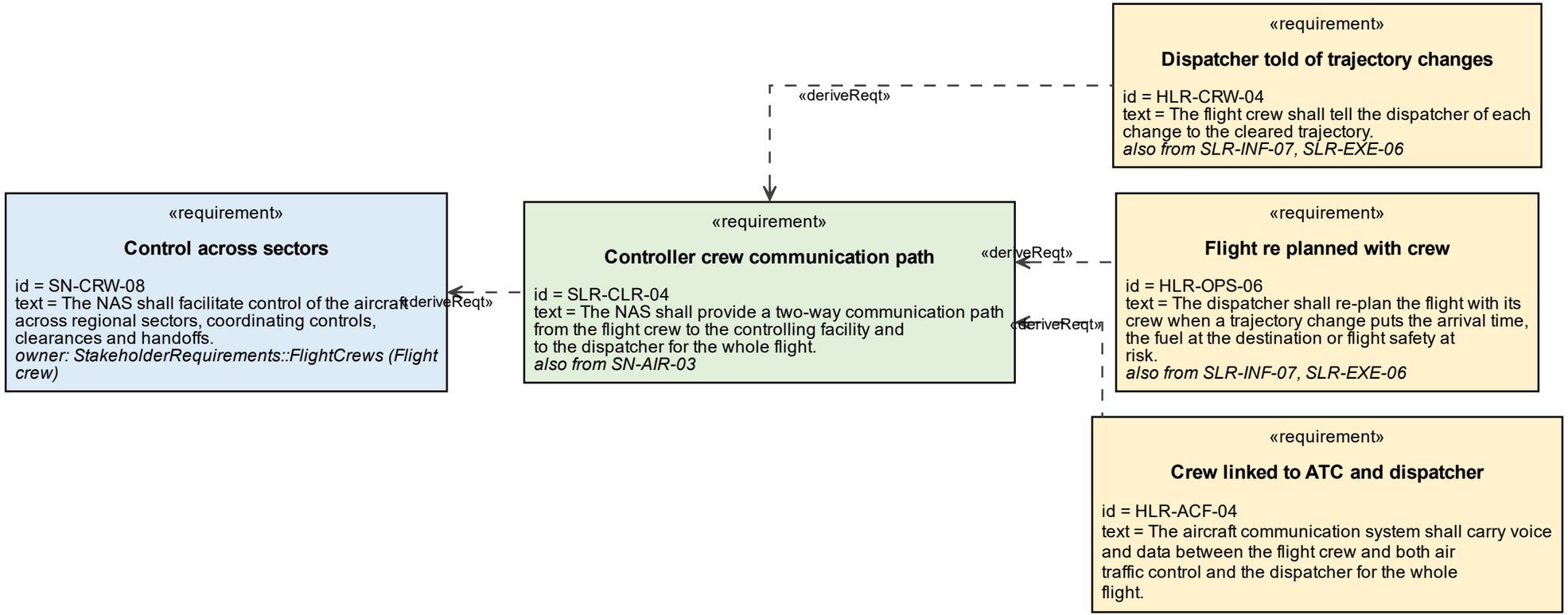
Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.



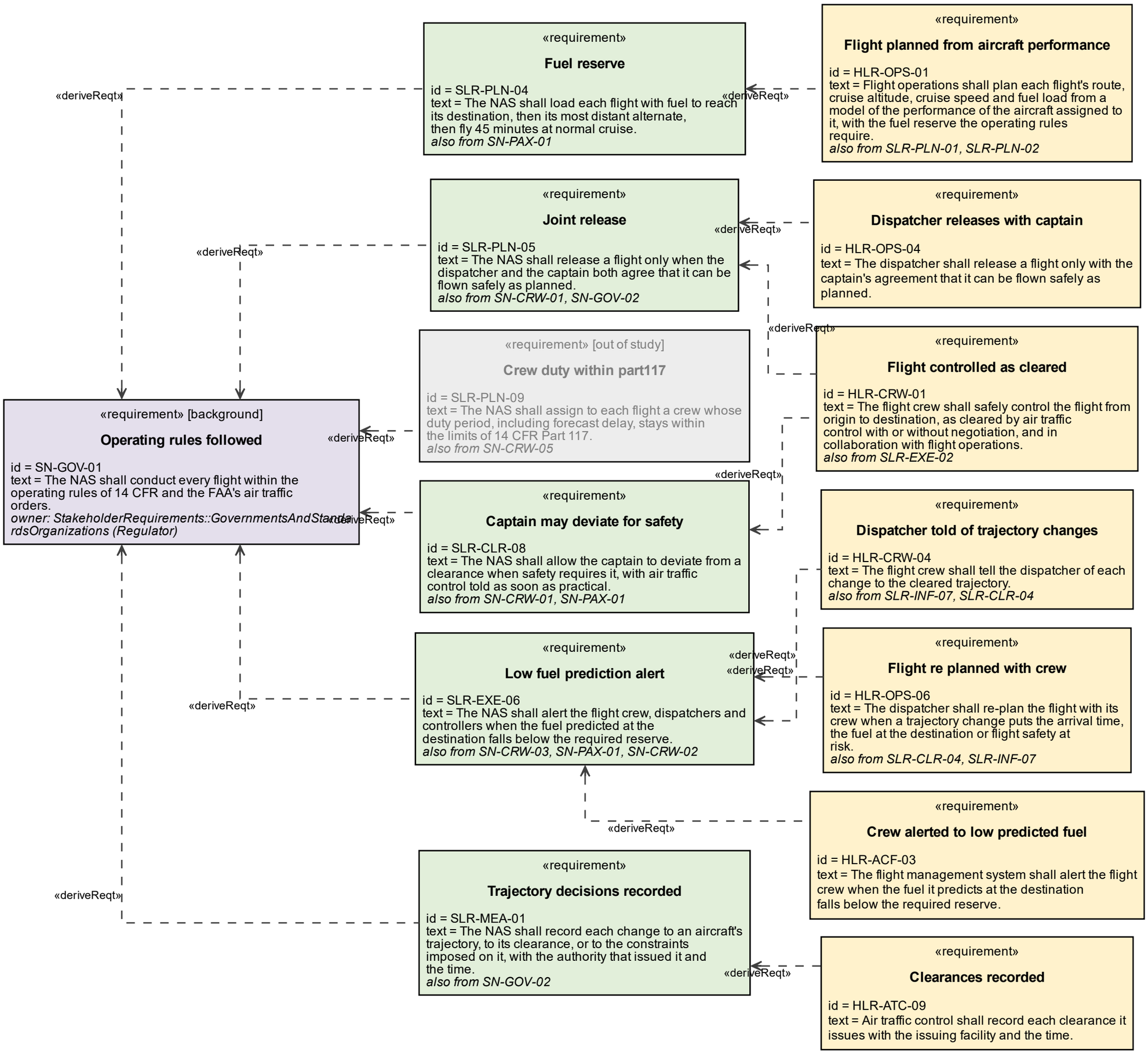
Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.



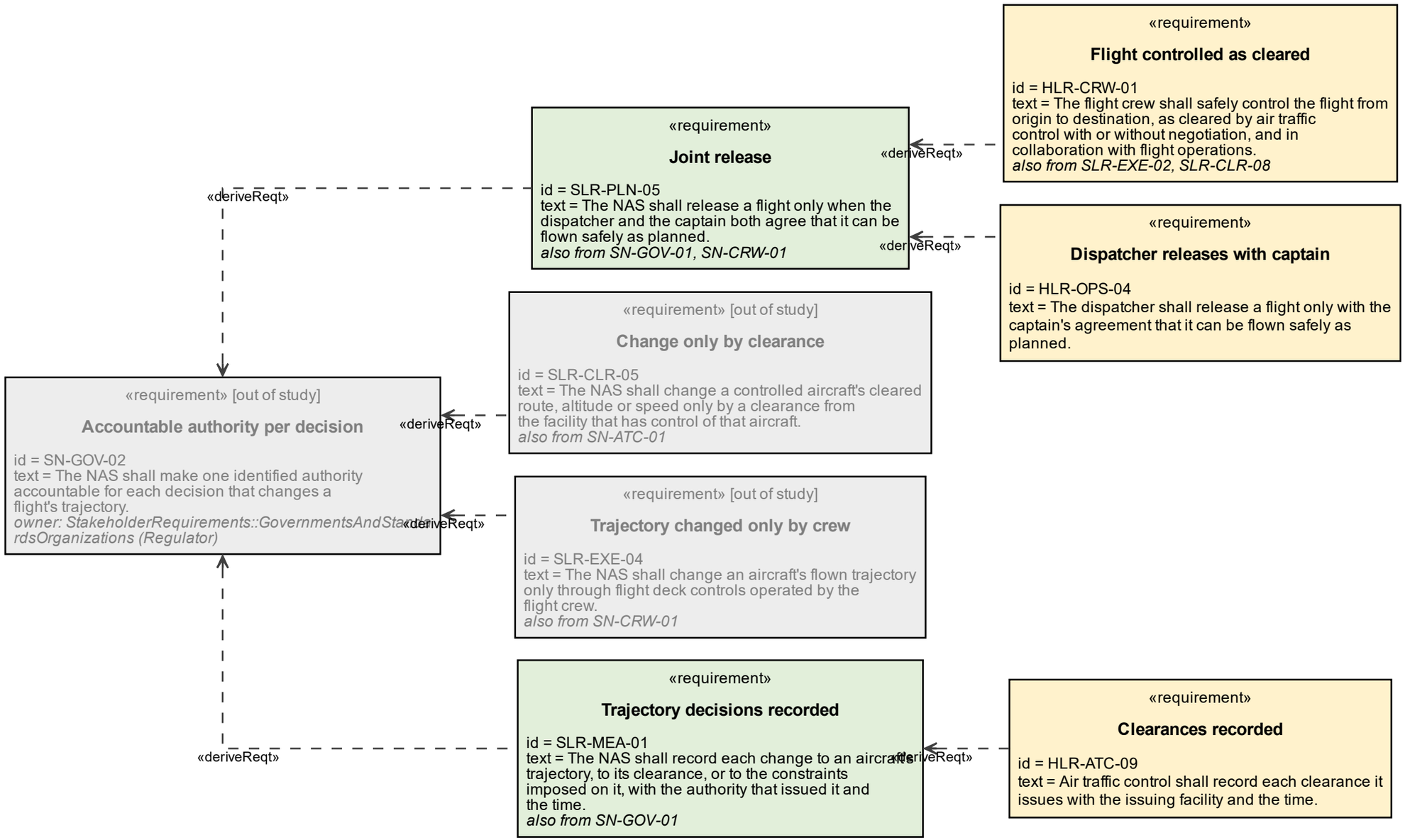
Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.



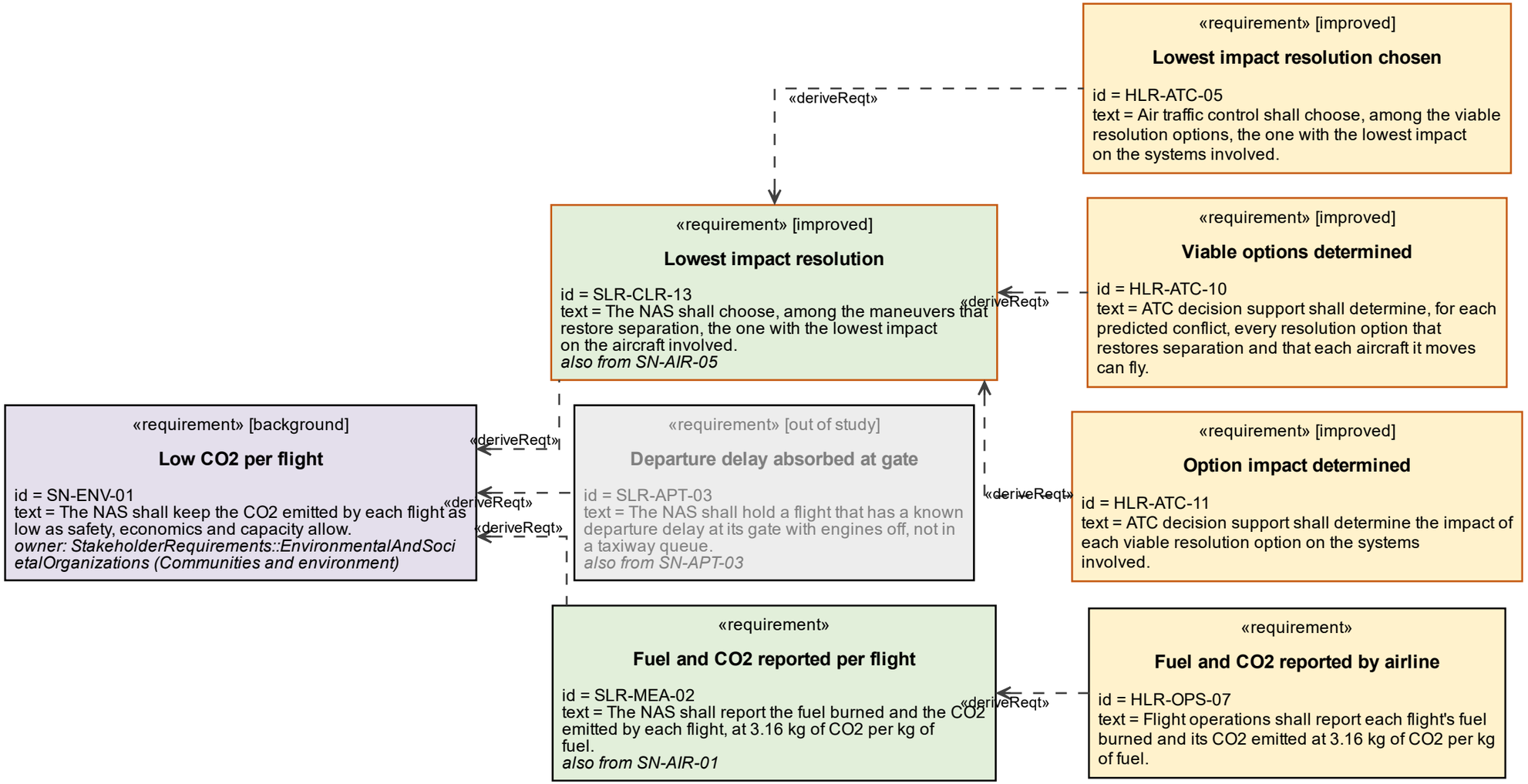
Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.



Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018). Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

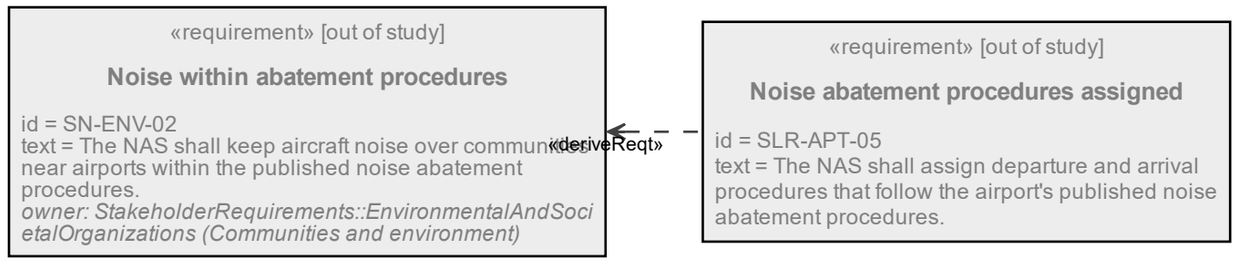


Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018). Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

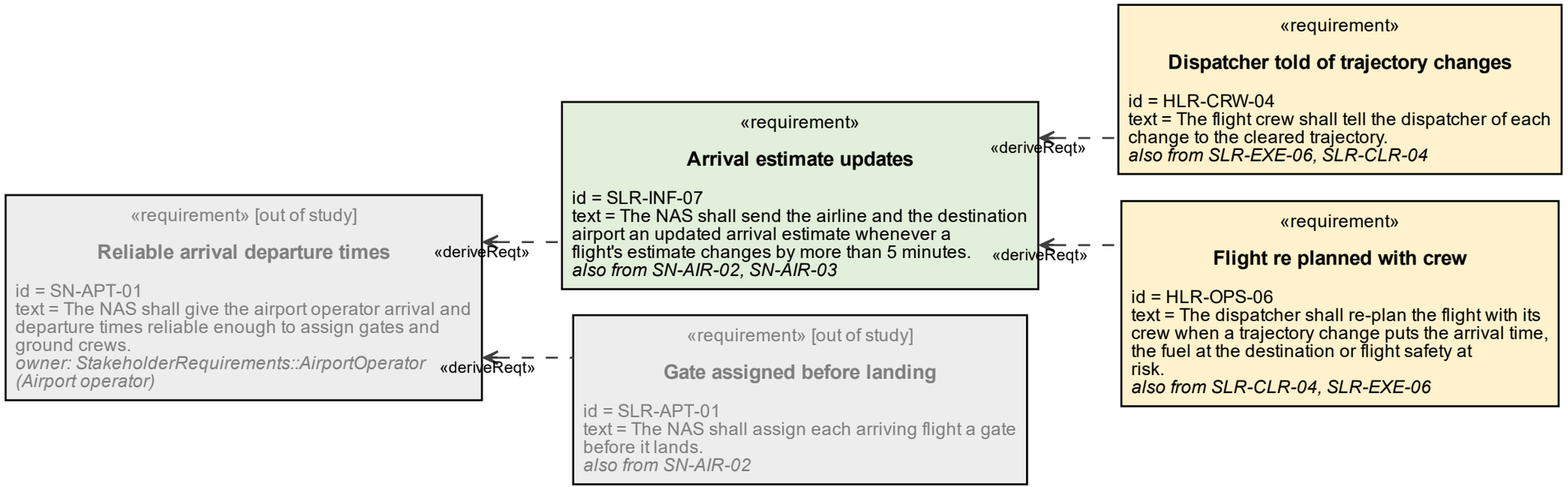


Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

No HLRs: the study is cut at the HLR tier (D-031).

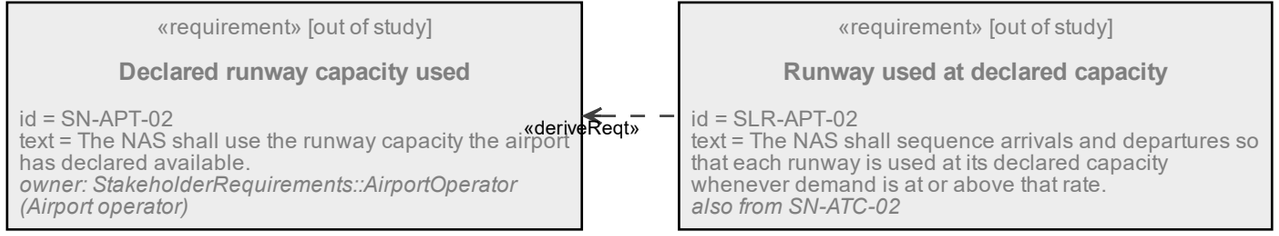


Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.



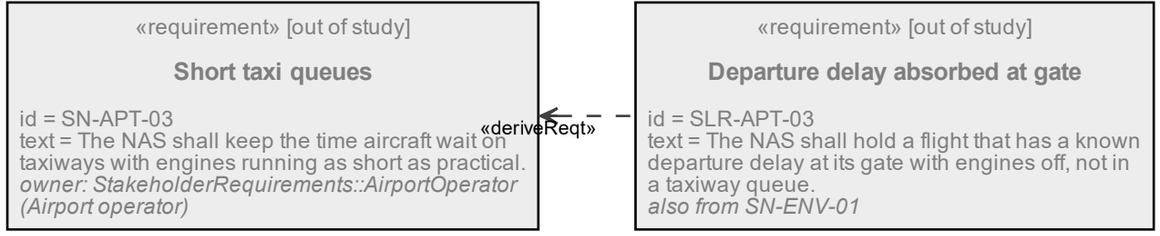
Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

No HLRs: the study is cut at the HLR tier (D-031).

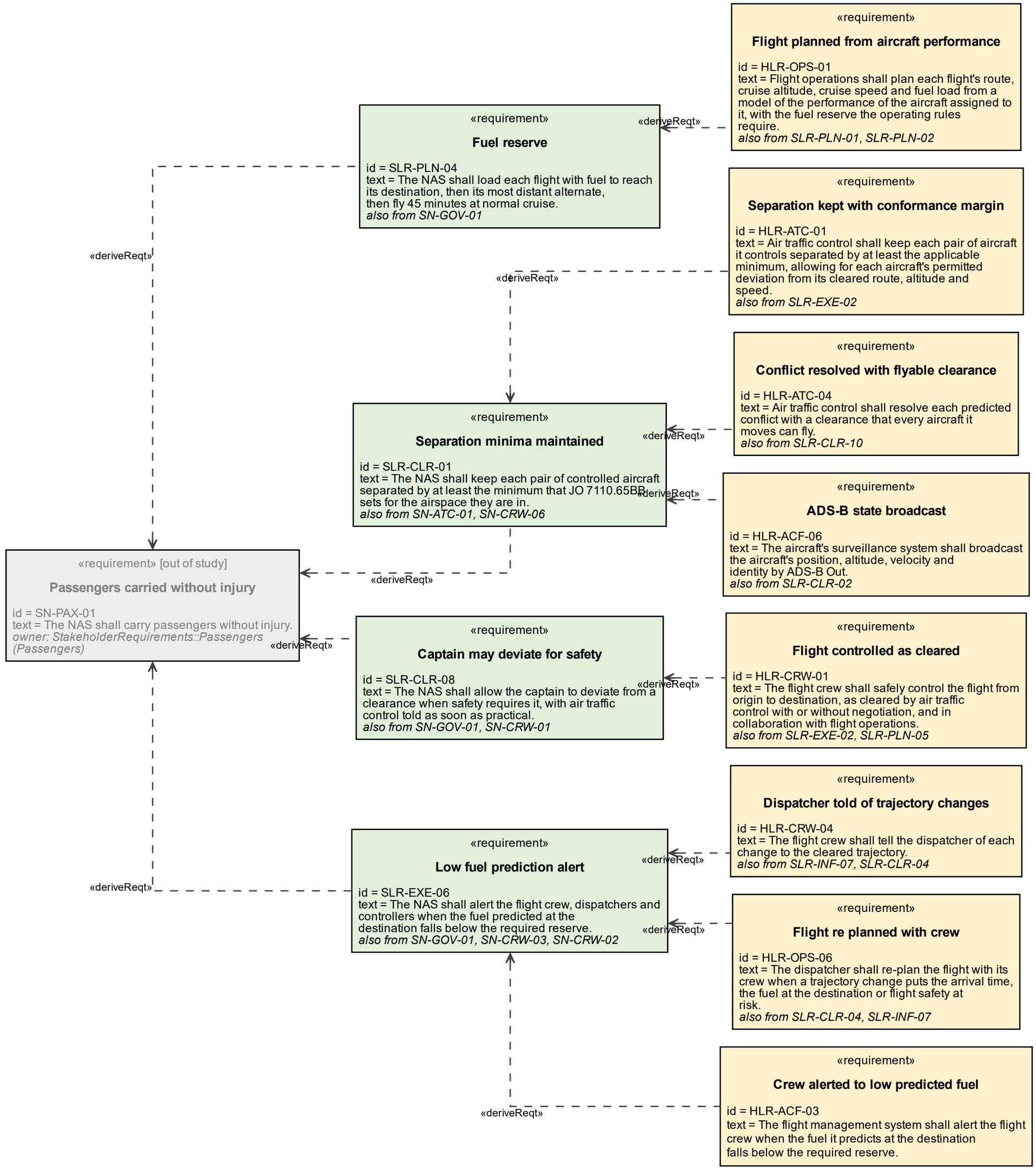


Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

No HLRs: the study is cut at the HLR tier (D-031).

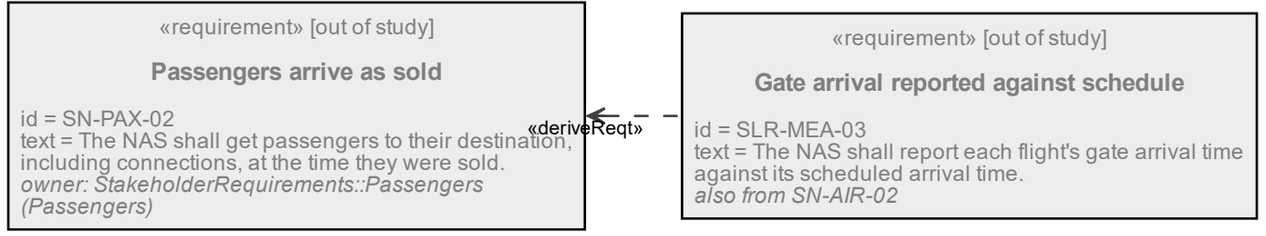


Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

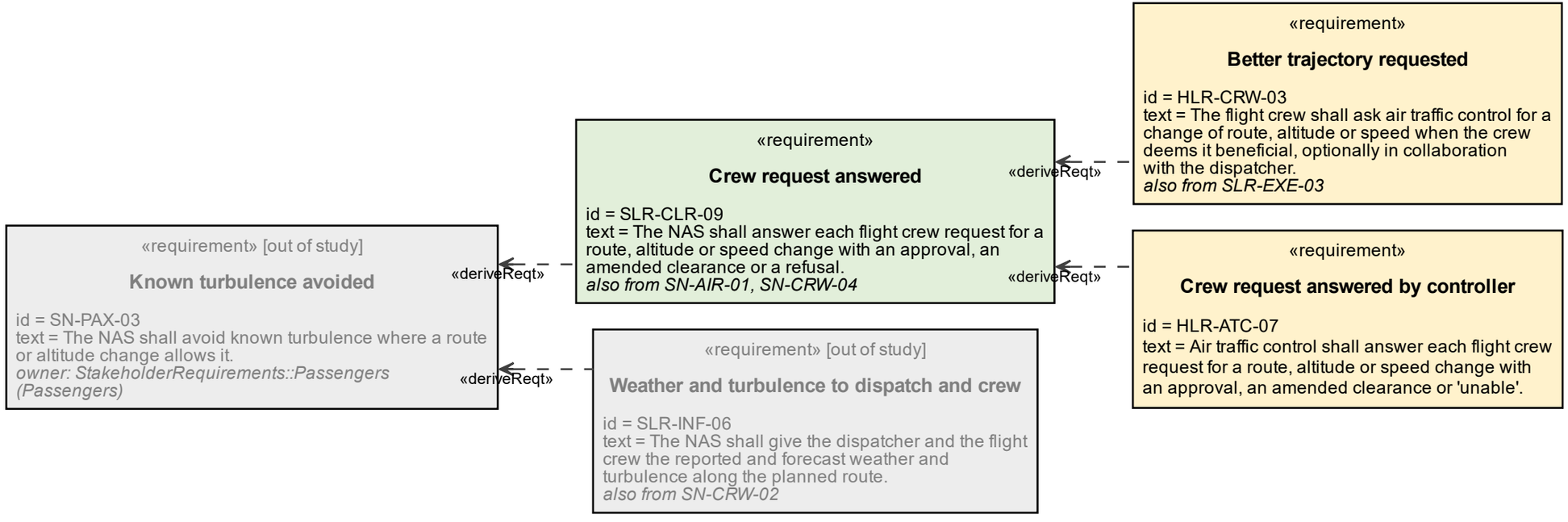


Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.

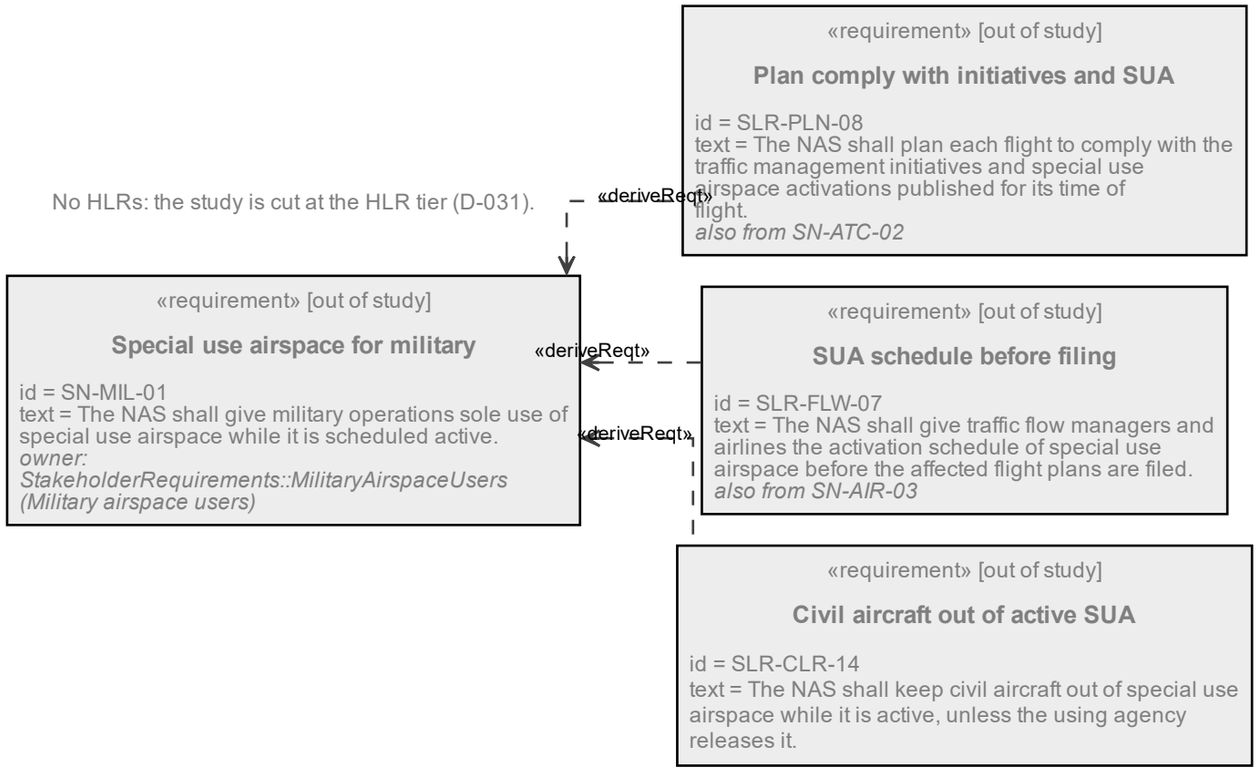
No HLRs: the study is cut at the HLR tier (D-031).



Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.



Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018). Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.



Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
Generated 2026-11-05 18:44 from commit a444913 (plus any uncommitted edits) by tools/export_requirements_diagram.py.